Practice Set 3 — Library Database

Q1. INNER JOIN members and loans. Predict the row count before running.

```
37 •  SELECT members.mname, loans.book_title, loans.fine
38    FROM members
39    INNER JOIN loans
40    ON members.member_id = loans.member_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

mname	book_title	fine
Ram	Python Basics	20
Ram	SQL Guide	60
Ram	Data Science	90
Sita	Database System	30
Sita	OOP in Python	80
Hari	Computer Network	40

Result 1

13 16:31:50 SELECT members.mname, loans.book_title, loans.fine FROM members INNER JOI... 8 row(s) returned

Q2. LEFT JOIN showing all members with their books. Who shows NULL and why?

```
43 •  SELECT members.mname, loans.book_title, loans.fine
44    FROM members
45    LEFT JOIN loans
46    ON members.member_id = loans.member_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

mname	book_title	fine
Ram	Python Basics	20
Ram	SQL Guide	60
Ram	Data Science	90
Sita	Database System	30
Sita	OOP in Python	80
Hari	Computer Network	40
Gita	Web Development	100
Gita	Java Programming	70
Nabin	NULL	NULL

14 16:33:31 SELECT members.mname, loans.book_title, loans.fine FROM members LEFT JOIN l... 9 row(s) returned

Q3. Swap the two tables in the LEFT JOIN. Which earlier join does it now behave like?

```
49 •  SELECT members.mname, loans.book_title
50    FROM loans
51    LEFT JOIN members
52    ON members.member_id = loans.member_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

mname	book_title
Ram	Python Basics
Ram	SQL Guide
Sita	Database System
Sita	OOP in Python
Hari	Computer Network
Gita	Web Development
Gita	Java Programming
Ram	Data Science

15 16:35:38 SELECT members.mname, loans.book_title FROM loans LEFT JOIN members ON ... 8 row(s) returned

Q4. RIGHT JOIN from members. What appears, and is any row NULL? Why not?

```
55 •  SELECT members.mname, loans.book_title, loans.fine
56    FROM loans
57    RIGHT JOIN members
58    ON members.member_id = loans.member_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

mname	book_title	fine
Ram	Python Basics	20
Ram	SQL Guide	60
Ram	Data Science	90
Sita	Database System	30
Sita	OOP in Python	80
Hari	Computer Network	40
Gita	Web Development	100
Gita	Java Programming	70
Nabin	NULL	NULL

16 16:36:57 SELECT members.mname, loans.book_title, loans.fine FROM loans RIGHT JOIN m... 9 row(s) returned

Q5. Write the FULL JOIN using UNION. How many rows?

```sql
SELECT members.member_id, members.mname,
       loans.book_title, loans.fine
FROM members
LEFT JOIN loans
ON members.member_id = loans.member_id

UNION

SELECT members.member_id, members.mname,
       loans.book_title, loans.fine
FROM members
RIGHT JOIN loans
ON members.member_id = loans.member_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

member_id	mname	book_title	fine
1	Ram	Python Basics	20
1	Ram	SQL Guide	60
1	Ram	Data Science	90
2	Sita	Database System	30
2	Sita	OOP in Python	80
3	Hari	Computer Network	40
4	Gita	Web Development	100

Result 5

17 16:39:05 SELECT members.member_id, members.mname, loans.book_title, loans.fine FR... 9 row(s) returned

Q6. Exclusive JOIN: members who borrowed nothing. Which IS NULL check?

```
76 •  SELECT members.member_id, members.mname, members.city
77    FROM members
78    LEFT JOIN loans
79    ON members.member_id = loans.member_id
80    WHERE loans.loan_id IS NULL;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

member_id	mname	city
5	Nabin	Chitwan

Result 6

18 16:40:42 SELECT members.member_id, members.mname, members.city FROM members LEF... 1 row(s) returned

Q7. Show member names with book titles only for fines above 50 (JOIN + WHERE).

```
83 •  SELECT members.mname, loans.book_title, loans.fine
84    FROM members
85    INNER JOIN loans
86    ON members.member_id = loans.member_id
87    WHERE loans.fine > 50;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

mname	book_title	fine
Ram	SQL Guide	60
Sita	OOP in Python	80
Gita	Web Development	100
Gita	Java Programming	70
Ram	Data Science	90

Result 7

19 16:41:56 SELECT members.mname, loans.book_title, loans.fine FROM members INNER JOI... 5 row(s) returned

Q8. Delete one loan, run the INNER JOIN again. Which rows disappeared?

```sql
SELECT members.mname, loans.book_title, loans.fine
FROM members
INNER JOIN loans
ON members.member_id = loans.member_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

mname	book_title	fine
Ram	Python Basics	20
Ram	SQL Guide	60
Sita	Database System	30
Sita	OOP in Python	80
Hari	Computer Network	40
Gita	Web Development	100
Gita	Java Programming	70
Ram	Data Science	90

Result 8

20 16:43:14 SELECT members.mname, loans.book_title, loans.fine FROM members INNER JOI... 8 row(s) returned

Q9. In your own words: when do you choose LEFT over INNER? Give a library example.

```sql
SELECT members.mname, loans.book_title
FROM members
LEFT JOIN loans
ON members.member_id = loans.member_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

mname	book_title	fine
Ram	Python Basics	20
Ram	SQL Guide	60
Sita	Database System	30
Sita	OOP in Python	80
Hari	Computer Network	40
Gita	Web Development	100
Gita	Java Programming	70
Ram	Data Science	90

Result 9

21 16:44:54 SELECT members.mname, loans.book_title, loans.fine FROM members INNER JOI... 8 row(s) returned

Q10. Bonus SELF JOIN: add a referred_by column to members (who invited whom) and list inviter with invited.

```sql
ALTER TABLE members
ADD referred_by INT;

UPDATE members
SET referred_by = 1
WHERE member_id IN (2, 3);

UPDATE members
SET referred_by = 2
WHERE member_id = 4;

SELECT inviter.mname AS inviter,
       invited.mname AS invited
FROM members AS invited
INNER JOIN members AS inviter
ON invited.referred_by = inviter.member_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

inviter	invited
Ram	Sita
Ram	Hari
Sita	Gita

Result 10

22	16:46:34	ALTER TABLE members ADD referred_by INT	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
23	16:46:38	UPDATE members SET referred_by = 1 WHERE member_id IN (2, 3)	2 row(s) affected Rows matched: 2 Changed: 2 Warnings: 0
24	16:46:42	UPDATE members SET referred_by = 2 WHERE member_id = 4	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
25	16:46:47	SELECT inviter.mname AS inviter, invited.mname AS invited FROM members A...	3 row(s) returned